

# ASTRA: Adaptive Satellite Task and Resource Allocator for LEO Earth Observation

Mohamed Habib Abid\*, Mariam Neffeti†

\*ISI El Manar (Higher Institute of Computer Science)

Tunisia

Email: mohamedhabib.abid@etudiant-isi.utm.tn

†INSAT (National Institute of Applied Science and Technology)

Tunisia

Email: mariam.neffeti@insat.ucar.tn

**Abstract**—Small Earth-observation (EO) satellites produce multispectral and synthetic-aperture-radar (SAR) payloads under intermittent ground contact and constrained battery, storage, compute, and thermal resources. We present ASTRA (Adaptive Satellite Task and Resource Allocator), a research prototype for IASTAM 6.0’s “Process or Transmit?” challenge. In the current simulator, we compare transmit-all, greedy-edge processing, a threshold heuristic, and one MaskablePPO policy. We evaluate each over five paired held-out scenarios of four 90-minute orbits, using decision quality, task completion, energy, latency, memory occupancy, single-event-upset (SEU), and safe-mode metrics. The threshold heuristic attains a mean delivered-utility ratio of 0.0286 and task completion rate of 1.60%; greedy-edge completes 5.17% of generated payloads but attains a utility ratio of 0.00470. The single-seed PPO checkpoint delivers no tasks; its 206.0 kJ mean energy use is lower than the processing-heavy policies but higher than transmit-all, consistent with an inactive-but-not-idle policy rather than an efficiency gain. This PPO result is diagnostic rather than a candidate operational policy and motivates the reward and training-time investigation in Section IV-B. A separate two-orbit trace reaches 52.48 degrees Celsius despite a 50-degree compute-throttling threshold. These synthetic results are preliminary; the thermal threshold is not a hard safety cap, and PPO was trained only briefly. We provide reproducible code and episode-level outputs; Lyapunov and oracle evaluations remain future work.

**Index Terms**—orbital edge computing, satellite task scheduling, reinforcement learning, constrained optimization, Low Earth Orbit.

## I. INTRODUCTION

The 2026 edition of the IEEE IAS Tunisia Annual Meeting (IASTAM 6.0) Technical Challenge, organized in collaboration with the Tunisian Space Association (TUNSA), addresses an operational bottleneck in Earth-observation data handling. Orbital edge computing has drawn attention as a way to move selected processing closer to sensors [1], [8]. Google’s Project Suncatcher is a proposed space-based AI infrastructure concept, not an operational orbital testbed [8].

Within this domain, Problem 1 (“Process or Transmit?”) targets an acute bottleneck in Earth Observation (EO): small satellites continuously capture high-resolution optical and

Synthetic Aperture Radar (SAR) payloads (generating multi-megabyte payloads every 15 s), yet operate under three binding physical constraints:

- 1) **Battery Floor:** The simulator models a 100 Wh battery and clamps its state at a 30% minimum depth-of-discharge floor.
- 2) **Thermal Constraint:** Vacuum requires radiative heat rejection. The current simulator uses a first-order thermal proxy and throttles compute at 50°C; this is not a hard cap, and the measured trace reaches 52.48°C.
- 3) **Severe Downlink Bandwidth Choke:** Line-of-sight contact with ground stations is limited to short windows (120–180 s per 90-minute orbit, about 2.8% mean contact duty cycle) with an X-band rate of 20.0 Mbps (2.5 MB/s).

For every captured payload, the onboard system must choose among four actions: PROCESS\_NOW (compression vs. inference), STORE\_QUEUE (time-shifted execution in Mass Memory Unit, MMU), TRANSMIT\_RAW (downlink during line-of-sight passes), or DISCARD\_DROP (pruning low-utility or cloud-occluded data). The optimization objective is to maximize cumulative scientific and operational utility, which decays exponentially over latency.

### Contributions:

- We implement paired benchmarks for transmit-all, greedy-edge, the threshold heuristic, and an initial MaskablePPO policy.
- We report eight outcome and resource metrics over five four-orbit evaluation seeds, including per-policy uncertainty.
- We identify an observed temperature excursion above the code’s 50°C throttling threshold, a limitation that motivates thermal-model and safety-guard validation.

## II. RELATED WORK

### A. Orbital and Edge Computing Systems

Orbital edge computing work describes the potential for processing data near the sensor [1]. Trabant proposes a serverless architecture for multi-tenant orbital computing [2], while the Tiansuan BUPT-1 study reports a cloud-native satellite case study [3]. Broader surveys discuss AI in space [4] and onboard

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Equal contribution.

Earth-observation image processing [5]. FPGA accelerator research [6] and compute-placement analysis [7] provide context for the design space. The numerical workload, energy, and thermal parameters used here are simulator assumptions; they are not calibrated or copied as empirical operating points from these studies.

### B. Control and Scheduling Paradigms

Resource-constrained orbital scheduling is addressed via two dominant paradigms:

- 1) **Lyapunov Optimization & Drift-Plus-Penalty:** Prior work [10] motivates online scheduling; implementing and evaluating an orbital scheduler remains planned work.
- 2) **Constrained Reinforcement Learning (CMDP):** We train an initial MaskablePPO policy using the repository's Gymnasium environment and invalid-action masks; this one-seed result is exploratory.

### C. Added Value of This Work

Prior work motivates orbital edge processing and constrained scheduling. This interim study provides a reproducible comparison of two operational reference policies, the threshold heuristic, and one exploratory learned policy, and identifies model and learning gaps.

## III. METHODOLOGY

### A. Physical Models & Orbital Mechanics

The satellite orbital environment and hardware state at time  $t$  are formalized in Table I.

TABLE I  
PRIMARY SYSTEM AND STATE VARIABLES

| Symbol                                   | Physical Definition                        | Nominal Value / Range                                   |
|------------------------------------------|--------------------------------------------|---------------------------------------------------------|
| $\tau_{\text{orbit}}, \tau_{\text{sun}}$ | Orbit period / Sunlit duration             | 5400 s / 3300 s (90 / 55 min)                           |
| $P_{\text{harvest}}(t)$                  | Solar power harvested                      | 0.0 W (ecl) to 30.0 W peak                              |
| $\text{SoC}(t)$                          | Battery State-of-Charge                    | Floor $\geq 0.30$ , Cap: 100.0 Wh                       |
| $T_{\text{chip}}$                        | Processor temperature                      | Ambient 0°C, throttle threshold 50.0°C (not a hard cap) |
| $G(t)$                                   | Ground station contact flag                | {0, 1}, 120–180 s pass (about 2.8% mean duty)           |
| $P_{\text{downlink}}$                    | RF communication speed                     | 20.0 MBps = 2.5 MB/s (15.0 W)                           |
| $P_{\text{seu}}$                         | Simulated per-second SEU-proxy probability | $10^{-5}$ RAM-reset event probability                   |
| $M_{\text{mmu}}, M_{\text{ram}}$         | Mass Memory / RAM                          | 32 000 MB / 4 000 MB                                    |

1) **Power and Battery Dynamics:** The simulator uses a 55-minute sunlight interval followed by a 35-minute eclipse. In sunlight, harvested power follows a sine-shaped profile from a 22.5 W base to a 30 W peak, with 30 s cosine tapers at sunlight boundaries; eclipse harvesting is zero.

Battery energy is updated by

$$E_{\text{batt}}(t+1) = \min(E_{\text{cap}}, E_{\text{batt}}(t) + [P_{\text{harvest}}(t) - P_{\text{draw}}(t)]\Delta t), \quad (1)$$

where  $P_{\text{draw}} = P_{\text{base}} + P_{\text{compute}} + P_{\text{comm}}$ ,  $P_{\text{base}} = 5$  W, and radio draw is 15 W during downlink. The implementation has no charge/discharge efficiency; it clamps battery energy to the 30% floor on brownout and exits safe mode above 1.05 times that floor.

2) **Lumped Thermal Proxy:** The simulator uses a first-order thermal proxy rather than a Stefan–Boltzmann radiation model:

$$T_{\text{chip}}(t+1) = T_{\text{chip}}(t) + 0.15P_{\text{draw}}(t) - 0.10(T_{\text{chip}}(t) - T_{\text{amb}}). \quad (2)$$

This one-second update uses watts of total bus, compute, and communication draw, with  $T_{\text{amb}} = 0^\circ\text{C}$ . Compute pauses at or above  $50^\circ\text{C}$ , but this threshold is not a hard cap; the seed-42 two-orbit trajectory peaks at  $52.48^\circ\text{C}$ .

### B. Workload, Queueing, and Utility Formulation

Payloads  $k$  arrive periodically every  $\Delta t_{\text{gen}} = 15$  s across dual modalities:

- **Optical Multispectral (70%):** Size  $S_k \in [5, 15]$  MB, initial utility  $U_{0,k} \in [10, 100]$ .
- **Synthetic Aperture Radar (30%):** Size  $S_k \in [100, 300]$  MB, initial utility  $U_{0,k} \in [10, 100]$ .

Payload utility decays exponentially over latency ( $t - t_{\text{capture}}$ ):

$$U_k(t) = U_{0,k} \cdot \exp(-\alpha_k(t - t_{\text{capture}})) \quad (3)$$

with decay parameter  $\alpha_k \in [10^{-4}, 10^{-3}] \text{ s}^{-1}$ .

Compute actions transform payload properties:

- **Compression:** Duration 2 s,  $P_{\text{comp}} = 10$  W, size factor  $C_{\text{ratio}} = 0.30$ , utility retention  $\rho_{\text{val}} = 0.90$ .
- **Inference:** Duration 5 s (with 2 s wake penalty),  $P_{\text{inf}} = 20$  W (30 W wake), size factor  $C_{\text{ratio}} = 0.001$ , utility retention  $\rho_{\text{val}} = 0.50$ .

### C. Decision Engine & Solution Schedulers

For each payload  $k$ , the agent selects an action  $a_k \in \mathcal{A}$ , where:

$$\mathcal{A} = \{\text{PROCESS\_COMPRESS, PROCESS\_INFER, DOWNLINK\_RAW, STORE\_QUEUE, DISCARD\_DROP}\} \quad (4)$$

- 1) **Transmit-all:** Sends the oldest raw MMU payload during contact, without onboard processing or voluntary drops.
- 2) **Greedy-edge:** Processes the oldest eligible payload when queue capacity permits, then downlinks processed outputs during contact.
- 3) **Threshold heuristic:** Uses contact-aware downlink, SoC/temperature-gated processing, and stale-data eviction under storage pressure.
- 4) **MaskablePPO:** Uses normalized telemetry and top- $K$  payload metadata with invalid-action masks. This initial learned policy is exploratory, not tuned or safety-certified.
- 5) **Planned methods:** Lyapunov scheduling and a non-causal MILP oracle are not implemented.

TABLE II  
PAIRED POLICY COMPARISON ACROSS FIVE FOUR-ORBIT EPISODES (MEAN; SAMPLE SD; 95% CI)

| Metric                     | Transmit-all                       | Greedy-edge                        | Heuristic                         | MaskablePPO (seed 7)              |
|----------------------------|------------------------------------|------------------------------------|-----------------------------------|-----------------------------------|
| Decision quality           | .00183; .00115; [.00040,.00326]    | .00470; .00105; [.00339,.00601]    | .02860; .00142; [.02683,.03036]   | 0; 0; [0,0]                       |
| Completed-task rate        | .01569; .00212; [.01307,.01832]    | .05167; .00805; [.04168,.06166]    | .01597; .00049; [.01536,.01658]   | 0; 0; [0,0]                       |
| Energy (kJ)                | 117.150; .340; [116.728,117.571]   | 290.882; 1.761; [288.695,293.068]  | 290.882; 1.761; [288.695,293.068] | 206.012; 4.782; [200.075,211.949] |
| Delivery latency (s)       | 9658.52;1995.69;[7180.95,12136.10] | 9922.94;1289.76;[8321.74,11524.14] | 753.41;121.36;[602.75,904.07]     | N/A (no delivery)                 |
| MMU utilization (%)        | 82.81;1.96;[80.39,85.24]           | 38.16;1.91;[35.79,40.54]           | 37.96;1.90;[35.60,40.32]          | 48.77;3.63;[44.27,53.28]          |
| RAM utilization (%)        | 0.0;[0,0]                          | .173;.003;[.169,.177]              | .173;.003;[.169,.177]             | .439;.036;[.394,.484]             |
| SEU events / episode       | 0.0;[0,0]                          | 0.0;[0,0]                          | 0.0;[0,0]                         | 0.0;[0,0]                         |
| Safe-mode events / episode | 0.0;[0,0]                          | 0.0;[0,0]                          | 0.0;[0,0]                         | 0.0;[0,0]                         |

Note: Greedy-edge and the heuristic have identical per-seed energy totals in this run. Direct reruns confirm equal aggregate base, compute, and radio draw for each paired seed; the equality is specific to this workload and simulator run.

#### IV. EXPERIMENTS AND RESULTS

##### A. Experimental Setup

Four policies share the same simulator configuration, four-orbit horizon (21,600 one-second steps), and five paired evaluation seeds (1001–1005); the heuristic is the reference. MaskablePPO was trained with seed 7 for 100,000 requested timesteps, with evaluation seeds disjoint from training. Table II reports episode means, sample standard deviations, and two-sided 95% Student- $t$  intervals ( $n = 5$ ). Decision quality divides delivered utility by the snapshot sum of generated payload base values. Memory use is time-averaged occupancy; energy includes modeled base, compute, and communication draw; latency averages completed deliveries only. Since only one PPO training seed was evaluated, its result does not measure training-seed variability. PPO is included as a diagnostic baseline for reward and training investigation, not as a candidate operational policy.

##### B. Preliminary Results

The first PPO checkpoint completed no payload in the five held-out episodes. Its mean energy use (206.0 kJ) is below greedy-edge and the heuristic (290.9 kJ) but above transmit-all (117.2 kJ); with zero delivered tasks, this reflects inactivity rather than energy efficiency. This diagnostic result motivates reward and training investigation; PPO is not a candidate operational policy in its present state. Greedy-edge and the heuristic also have exactly equal per-seed energy in this run. Direct reruns decompose each total into identical base, compute, and radio energy for each paired seed, although payload choices and delivery outcomes differ. No SEUs occurred in these five evaluation episodes, which is too few to assess rare-event recovery. Across a separate two-orbit seed-42 trace, SoC ranged from 92.37% to 100%, while temperature ranged from 0.75°C to 52.48°C. The 50°C threshold throttles processing but does not bound simulated temperature.

Greedy-edge’s larger completion fraction does not imply higher delivered utility: the utility ratio is lower than for the threshold heuristic. Its matching energy reflects the same aggregate modeled power draw on these five seeds, not identical decisions, and should not be generalized beyond this run. The paired results are preliminary ( $n = 5$ ), and the PPO diagnostic uses one training seed. Transmit-all and greedy-edge are operational reference policies, not optimized

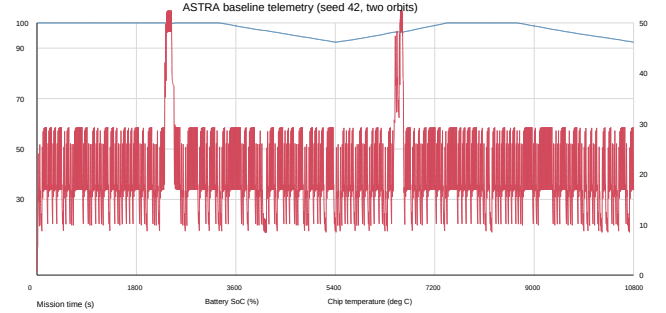


Fig. 1. Heuristic trajectory over two orbits (seed 42). Temperature exceeds the 50°C throttle threshold.

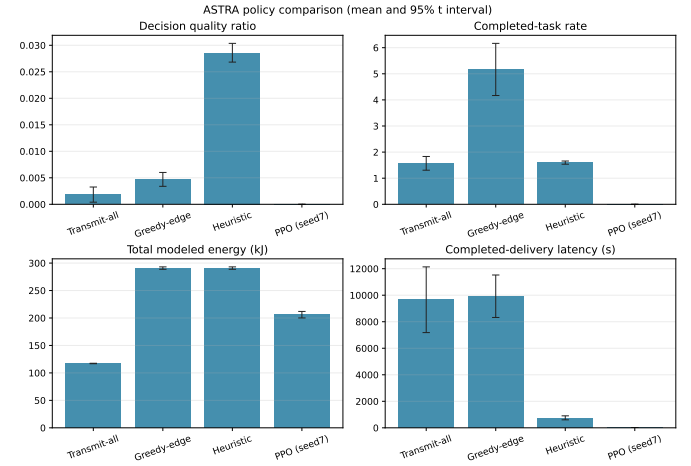


Fig. 2. Policy comparison; bars show means and error bars show 95%  $t$  intervals. PPO completes no tasks in this run.

implementations. Lyapunov scheduling and the MILP oracle remain future work.

##### C. Phase 3 Validation Plan

Next, diagnose sparse-reward learning and improve PPO training using multiple independent training seeds and validation episodes, then evaluate a selected checkpoint on a fresh test set. Further work includes Lyapunov scheduling, a non-causal MILP oracle, and stress tests for SEU bursts, battery degradation, and missed contacts.

## V. CONCLUSION

We implemented a paired comparison of transmit-all, greedy-edge, the threshold heuristic, and an initial MaskablePPO diagnostic baseline in one four-orbit simulator benchmark. The heuristic achieved the highest delivered-utility ratio (0.0286), while greedy-edge completed more tasks (5.17%). The single-seed PPO policy delivered no tasks; its 206.0 kJ energy use is below the processing-heavy policies but above transmit-all (117.2 kJ), so it is neither evidence of an efficiency gain nor a candidate operational policy. The reproduced equality in greedy-edge and heuristic energy reflects their identical aggregate modeled power draw on these seeds, not identical decisions. The simplified thermal model can exceed its throttle threshold; all results are synthetic, not flight-validated. Follow-up work should address these limits and add multiple PPO seeds, Lyapunov scheduling, and an oracle benchmark.

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